

# Лабораторная работа №5

## Презентация

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# Раздел 1

## Информация

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## Раздел 2

### Выполнение лабораторной работы

# Запускаем наш сервер

Попробуйте новую кроссплатформенную оболочку PowerShell (<https://aka.ms/pscore6>)

```
PS C:\Users\Ванек> cd C:\work\iazubov\vagrant
PS C:\work\iazubov\vagrant> vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
    server: Adapter 1: nat
    server: Adapter 2: intnet
==> server: Forwarding ports...
    server: 22 (guest) => 2222 (host) (adapter 1)
==> server: Running 'pre-boot' VM customizations...
==> server: Booting VM...
==> server: Waiting for machine to boot. This may take a few minutes...
    server: SSH address: 127.0.0.1:2222
    server: SSH username: vagrant
    server: SSH auth method: password
==> server: Machine booted and ready!
==> server: Checking for guest additions in VM...
    server: No guest additions were detected on the base box for this VM! Guest
    server: additions are required for forwarded ports, shared folders, host only
    server: networking, and more. If SSH fails on this machine, please install
    server: the guest additions and repackage the box to continue.
    server:
    server: This is not an error message; everything may continue to work properly,
```

# Создаем каталог private

```
root@server:~# mkdir -p /etc/pki/tls/private
root@server:~# ln -s /etc/pki/tls/private /etc/ssl/private
root@server:~# cd /etc/pki/tls/private
root@server: /etc/pki/tls/private# █
```

## Сгенерируем ключ и сертификат

```
root@server:/etc/pki/tls/private# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.iazubov.net.key -out www.iazubov.net.crt
```

```
. . . + . + . . . . + + . . . . . + . . . . . + . . . + . . . . . ++++++
+++++++* . . . + . + . . . + . . . ++++++
++++++* . . . . . ++++++
```

[illegible]

You are about to be asked to enter information that will be incorporated into your certificate request.

What you are about to enter is what is called a Distinguished Name or a DN.

There are quite a few fields but you can leave some blank

For some fields there will be a default value.

If you enter '.', the field will be left blank.

Country Name (2 letter code) [XX]:RU

State or Province Name (full name) []:Russia

Locality Name (eq. city) [Default City]:Moscow

Organization Name (eg, company) [Default Company Ltd]:iazubov

Organizational Unit Name (eg, section) []:iazubov

Common Name (eq. your name or your server's hostname) [ ]: iazubov.net

Перейдем в каталог с конфигурационными файлами и отредактируем его.

```
GNU nano 8.1          www.iazubov.net.conf          Modified
<VirtualHost *:80>
ServerAdmin webmaster@iazubov.net
DocumentRoot /var/www/html/www.iazubov.net
ServerName www.iazubov.net
ServerAlias www.iazubov.net
ErrorLog logs/www.iazubov.net-error_log
CustomLog logs/www.iazubov.net-access_log common
RewriteEngine on
RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>
<IfModule mod_ssl.c>
<VirtualHost *:443>
SSLEngine on
ServerAdmin webmaster@iazubov.net
DocumentRoot /var/www/html/www.iazubov.net
ServerName www.iazubov.net
ServerAlias www.iazubov.net
ErrorLog logs/www.iazubov.net-error_log
CustomLog logs/www.iazubov.net-access_log common
```

## Внесем изменения в настройки межсетевого экрана на сервере

```

root@server:/etc/httpd/conf.d# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http ssh
root@server:/etc/httpd/conf.d# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client
amqp amqps anno-1602 anno-1800 apcupsd aseqnet audit ausweisapp2 bacula
bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoi
n bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-ex
porter ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit
collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dh
cpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker-registry dock
er-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finge
r foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replicat
ion freeipa-trust ftp galera ganglia-client ganglia-master git gitea gpsd gra
fana gre high-availability http http3 https ident imap imaps iperf2 iperf3 ip
fs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogon kpasswd kprop kshell kube-api kube-apiserver kube-cont
rol-plane kube-control-plane-secure kube-controller-manager kube-controller-m
anager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kub
e-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-t
ls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matri
x mdns memcache minecraft minidlna mndp mongodb mosh mountd mpd mqtt mqtt-tls
ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns n
etdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-
imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmpoxy pmwebapi pmweb
apis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-
dhcp proxy-http ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius
radsec rdp redis redis-sentinel rootd rpc-bind rquotd rsh rsyncd rtsp salt-m
aster samba samba-client samba-daemon samba-dc samba-dc-guest samba-dc-history collection sip sip

```

# Внесем изменения в настройки межсетевого экрана на сервере

```
root@server:/etc/httpd/conf.d# firewall-cmd --add-service=https
success
root@server:/etc/httpd/conf.d# firewall-cmd --add-service=https --permanent
success
root@server:/etc/httpd/conf.d# firewall-cmd --reload
```

# Перезапускаем сервер

success

```
root@server:/etc/httpd/conf.d# systemctl restart httpd
```

```
root@server:/etc/httpd/conf.d# █
```

# Проверка https



## Certificate

iazubov.net

### Subject Name

|                     |                     |
|---------------------|---------------------|
| Country             | RU                  |
| State/Province      | Russia              |
| Locality            | Moscow              |
| Organization        | iazubov             |
| Organizational Unit | iazubov             |
| Common Name         | iazubov.net         |
| Email Address       | iazubov@iazubov.net |

### Issuer Name

|                     |             |
|---------------------|-------------|
| Country             | RU          |
| State/Province      | Russia      |
| Locality            | Moscow      |
| Organization        | iazubov     |
| Organizational Unit | iazubov     |
| Common Name         | iazubov.net |

# Установим пакеты для работы с PHP

```
root@server:/etc/httpd/conf.d# dnf -y install php
Rocky Linux 10 - BaseOS                               15 kB/s | 4.3 kB      00:00
Rocky Linux 10 - BaseOS                               14 MB/s | 34 MB       00:02
```

# Редактируем файл



The screenshot shows a terminal window with a pink title bar. The title bar text is "root@server:/var/www/html/www.iazubov.net – sudo -i". Inside the terminal, the GNU nano 8.1 editor is open, editing the file "index.php". The editor's status bar at the top shows "GNU nano 8.1", "index.php", and "Modified". The code in the editor consists of three lines: "<?php" in green, "phpinfo();" in grey, and "?>" in green. A cursor is positioned at the end of the third line.

```
root@server:/var/www/html/www.iazubov.net – sudo -i
GNU nano 8.1 index.php Modified
<?php
phpinfo();
?>
```

# Права доступа, контекст безопасности и перезапуск сервера

```
root@server:/var/www/html/www.iazubov.net# chown -R apache:apache /var/www
root@server:/var/www/html/www.iazubov.net# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from uncon
fined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw
_t:s0
root@server:/var/www/html/www.iazubov.net# restorecon -vR /var/www
root@server:/var/www/html/www.iazubov.net# systemctl restart httpd
```

# Проверка php

rockylinux.org/ x mozilla.org/privacy/firefox x HTTP Server Test Page pov x server.iazubov.net/ x PHP 8.3.33 - phpinfo()

www.iazubov.net

## PHP Version 8.3.33



|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System                                  | Linux server.iazubov.net 6.12.0-211.55.1.el10_2.x86_64 #1 SMP PREEMPT_DYNAMIC Tue Sep 15 14:08:11 UTC 2026 x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Build Date                              | Jul 28 2026 17:56:10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Build System                            | Rocky Linux release 10.2 (Red Quartz)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Build Provider                          | Rocky Enterprise Software Foundation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Compiler                                | gcc (GCC) 14.3.1 20251022 (Red Hat 14.3.1-4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Architecture                            | x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Server API                              | FPM/FastCGI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Virtual Directory Support               | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Configuration File (php.ini) Path       | /etc                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Loaded Configuration File               | /etc/php.ini                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Scan this dir for additional .ini files | /etc/php.d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Additional .ini files parsed            | /etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmlreader.ini |
| PHP API                                 | 20230831                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| PHP Extension                           | 20230831                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Zend Extension                          | 420230831                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

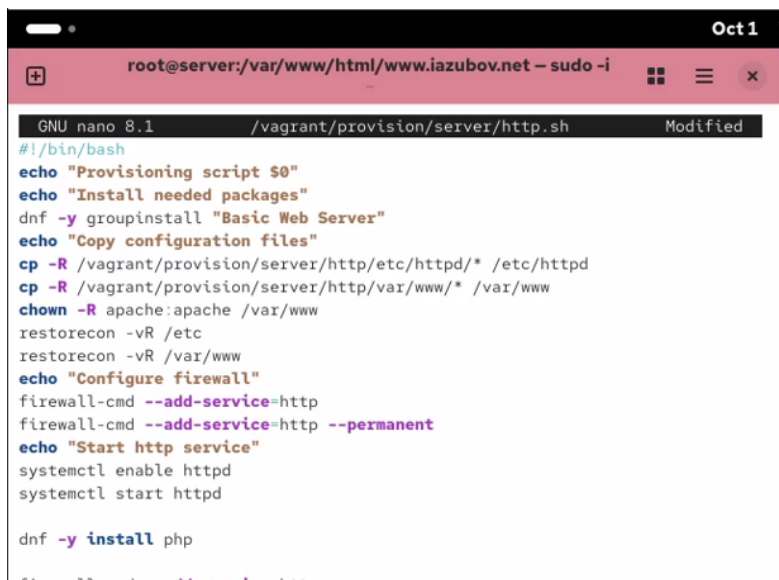
# Создание каталога для ключа и сертификата и копирование файлов

```
root@server:/var/www/html/www.iazubov.net# cp -R /etc/httpd/conf.d/* /vagrant
/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'
'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.iazubov
.net.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.user.ne
t.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'?

cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'?

cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.user.net.c
onf'?
root@server:/var/www/html/www.iazubov.net# cp -R /var/www/html/* /vagrant/pro
vision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/server.iazubov.net
/index.html'?
root@server:/var/www/html/www.iazubov.net#
root@server:/var/www/html/www.iazubov.net# mkdir -p /vagrant/provision/server
/http/etc/pki/tls/private
root@server:/var/www/html/www.iazubov.net# mkdir -p /vagrant/provision/server
/http/etc/pki/tls/certs
root@server:/var/www/html/www.iazubov.net# cp -R /etc/pki/tls/private/www.iaz
```

# Изменение скрипта http.sh



```
GNU nano 8.1 /vagrant/provision/server/http.sh Modified
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd

dnf -y install php
```